

PlantPulse — Incident Report

PlantPulse Demo Site — Process Unit 7 · Generated 2026-07-22 18:52

Incident	Slow Gas Leak Masked by Ventilation Failure
Zone	Z1
Window	2026-07-22T02:00:00 → 2026-07-22T04:00:00
Ground-truth severity	HIGH
Single-channel alarm would fire?	No — this is exactly the compound pattern PlantPulse exists to catch

Compound Risk at Peak

89 / 100

Confidence: 49%

Contributing Factors

- Sensor: +52%
- Weather: +36%
- Worker: +6%
- Equipment: +6%

Root Cause (Knowledge Graph)

Worker 05 enters Tank Farm (Z1) — Z1's standing entry permit P-01 is held by Worker 01, not Worker 05; TANK-01 shows Pressure falling, Gas Concentration rising; PIPE-01 shows Pressure falling; wind has dropped to near-calm, stalling dispersion.

SOP Response — SOP-GAS-01: Gas Leak / Toxic-Flammable Atmosphere Response

Source: OISD-STD-105 (Work Permit System) — permit suspension provisions; Factories Act 1948, Section 36 (Precautions against dangerous fumes, gases, etc.)

- 1 Suspend all active permits in the affected zone immediately — no new hot or cold work may begin
- 2 Sound the zone evacuation alert and account for all personnel via muster point roll call
- 3 Do not permit zone re-entry without appropriate respiratory PPE
- 4 Isolate the suspected source (valve/line) from outside the zone if it can be done safely
- 5 Increase ventilation or activate emergency exhaust where available to restore dispersion
- 6 Confirm gas detector readings return to the safe band before authorizing re-entry
- 7 Log the incident and complete root-cause review before any permit in the zone is reissued

Estimated risk reduction if followed: 7% (peak compound risk 89 → 83, computed by re-scoring a mitigated counterfactual through the same fusion model)

Detection Performance

Across 20 independently seeded evaluation days, PlantPulse detected this archetype in **100%** of instances with a median lead time of **138 minutes** over an extrapolated conventional single-sensor alarm. Source: engine/metrics/results.json.

Generated by PlantPulse from simulated scenario data (seed-reproducible). This is a demonstration report for the ET AI Hackathon 2.0 submission, not a record of a real industrial incident.